

Chapter 154 — Prostate Cancer Biomarkers

INTRODUCTION

- **Prostate cancer has a serum-based, clinically informative biomarker:** prostate-specific antigen (PSA), discovered 1979, used for detection, staging and monitoring.
- **Pre-PSA era:** diagnosis usually at an incurable stage, after an abnormal digital rectal examination (DRE) or symptomatic bone metastases. **PSA era:** most cancers detected by a modest PSA rise at an early, localised stage (T1c).
- **Stage migration to low-risk disease has paralleled the fall in prostate cancer mortality**, at least partly attributable to population-based PSA testing.
- **PSA is prostate specific, not cancer specific** → overlap between cancer and benign disease; even marked elevation may reflect inflammation or benign prostatic hyperplasia (BPH) → false positives → **excess negative biopsies and overdiagnosis of low-grade cancer**.
- **European Randomized Study of Screening for Prostate Cancer (ERSPC) and Göteborg-1:** PSA screening significantly reduced cancer-specific mortality, but PSA used as an isolated biopsy trigger produced a high negative-biopsy rate and overdiagnosis of Gleason score (GS) 6 / Grade Group (GG) 1 cancer.
- **US Preventive Services Task Force (USPSTF), 2012: Grade D recommendation against screening** on benefit-to-harm grounds.
- **Two advances have since mitigated screening harm:**
 - **Active surveillance (AS)** proven safe and oncologically effective for most low-risk (GS 6, GG 1) cancers → less overtreatment, blunting the harm of overdiagnosis.
 - **Guidelines advise secondary testing — imaging or biomarkers — before biopsy** when PSA is elevated; blood and urine biomarkers reduce unnecessary biopsies (negative, or GG 1 only).
- **Three clinical settings for biomarkers:**
 - **Early detection of clinically significant cancer** — before initial biopsy and before repeat biopsy (a particularly challenging group); a positive test selects men for imaging or biopsy.
 - **Risk stratification of newly diagnosed favourable-risk cancer** — AS versus curative therapy (radical prostatectomy or radiation).
 - **Post-treatment** — adjuvant or salvage decisions in men at high risk of recurrence or metastasis.
- **Molecular substrates:** DNA (gene) → messenger RNA (mRNA; transcript) → protein → products such as metabolites; **epigenetic modifications** regulating expression are of growing importance.
- **Guiding principle: cancer cells are biologically and molecularly distinct from benign cells;** identifying and quantifying these differences in tissue or body fluid is the lynchpin of biomarker discovery.
- **Second-generation, cancer-specific markers** — PSA isoforms (proPSA), RNA transcripts (PCA3), gene fusions (TMPRSS2:ERG) — improve detection of cancer and of **clinically significant cancer (GS ≥7 / GG ≥2)** relative to PSA. Presumed reason: **higher-grade tumours are larger, and larger tumours express more cancer-specific molecules**.
- **Third generation:** markers uniquely associated with aggressive, high-grade disease; only recently explored.

TABLE 154.1 EVOLUTION OF BIOMARKERS FOR DETECTION OF LOCALIZED PROSTATE CANCER (*Prensner et al., 2013*)

Generation	Biologic specificity	Example(s)
First	Prostatic epithelium	PSA
Second	Prostate cancer	PCA3; kallikrein-family proteins; ETS-related transcription factors and gene fusions
Third	High-grade prostate cancer	SChLAP1

- **What the table is for:** each generation narrows the biological target — organ → cancer → high-grade cancer.
- **Trap:** second-generation markers track high-grade cancer only **indirectly, through tumour volume**; only third-generation markers are intrinsically linked to grade.

Concept — *Biomarker development is a progressive narrowing of specificity: PSA fails because it tracks the organ rather than the disease, and every later marker is judged by how closely it tracks clinically significant cancer rather than any cancer.*

BLOOD-BASED BIOMARKERS

Prostate-Specific Antigen (PSA or hK3)

- Developed for **post-treatment monitoring**; now the guideline **first-line screening test** — affordable and accessible, but poorly cancer specific.

Kallikrein family.

- **PSA = human kallikrein 3 (hK3).** Family originally thought to be three — pancreatic/renal kallikrein (hK1), human kallikrein 2 (hK2), PSA (hK3); **now 15 proteins.**
- Genes on the **long arm of chromosome 19 (q13.2–q13.4)**; all encode **serine proteases**. Homology with PSA: **hK1 60%, hK2 about 78–80%.**
- **Fig. 154.1** (Yousef and Diamandis, 2001): kallikrein locus map, KLK1 to KLK15, with KLK2 (hK2) and KLK3 (PSA) adjacent.

Synthesis and function.

- **hK2 and PSA are both released as zymogens** (inactive proenzymes) from prostatic epithelium, present in semen and serum; both complex with α 2-macroglobulin and α 1-antichymotrypsin.
- **Processing sequence:** pre-pro-PSA with 17-amino acid (aa) leader → cleavage of the "pre" sequence → **proPSA, 244 aa, inactive, carrying a 7-aa "pro" leader** → **hK2 cleaves the pro leader** → **active PSA, 237 aa.**
- **Function: androgen-regulated protease that liquefies semen** by cleaving the gel-forming proteins semenogelin and fibronectin.
- Identified in the late 1970s; widespread urological use a decade later.
- Low-level ectopic expression: breast tissue, breast milk, adrenal and renal carcinomas. **Produced primarily by prostatic luminal epithelial cells** → highly organ specific.

Serum forms.

- **About 70% circulates bound — complexed PSA (cPSA)** — to α 1-antichymotrypsin (ACT), α 2-macroglobulin (A2M) or α 1-protease inhibitor (API); the remainder is **free PSA (fPSA).**
- **Fig. 154.2:** molecular forms — free PSA (proPSA and clipped forms, benign PSA, intact PSA) versus complexed PSA (PSA-ACT, PSA-A2M, PSA-API).

- **Fig. 154.3:** free PSA carries immunoreactive epitopes A–E; **ACT blocks epitope E; A2M blocks all epitopes**, making PSA-A2M difficult to measure.

TABLE 154.3 MOLECULAR DERIVATIVES OF PROSTATE-SPECIFIC ANTIGEN

PSA form	% of serum PSA	Immunodetectable	Protease activity
Complexed PSA (total)	60–95	—	—
PSA-ACT (predominant; 60–90% of cPSA)	60–90	Yes	Irreversibly inactivated
PSA-A2M	10–20	No — all epitopes masked	Some activity retained
PSA-API	1–5	Yes	—
Free PSA	5–45	Yes	Enzymatically inactive

- **Pattern:** "total PSA" on a standard assay = free PSA + PSA-ACT + PSA-API; **PSA-A2M is invisible** to most assays.
- **Trap:** A2M-bound PSA is the **active** complex yet the undetectable one; ACT-bound PSA is inactive yet detectable.

Androgen dependence.

- **PSA expression is strongly androgen dependent.** Immunohistochemical PSA in the prostate peaks bimodally — **0–6 months of age and after 10 years** — tracking testosterone; **serum PSA becomes detectable at puberty** with rising luteinising hormone and testosterone.

Causes of elevation.

- **Mechanism: disruption of glandular architecture** — loss of the basal-cell layer and basement-membrane barrier allows PSA to escape into the circulation.
- Causes: cancer, BPH, inflammation, **prostate manipulation** (massage, biopsy).
- **Prostate biopsy:** rise persists **4 weeks or more**.
- **Long-distance cycling** (>55 km): rise of about 10%.
- **Ejaculation:** most studies show no significant change; **in men ≥50, a transient rise, back to baseline within 24 hours** → ask about sexual activity within 24 hours of a new elevation; if present, abstain before repeating.
- **Prostate disease (cancer, BPH, prostatitis) is by far the most important factor; BPH and prostatitis are the primary confounders** of PSA screening.
- **Per cell, cancer expresses less PSA than benign tissue** — but cellular expression ≠ serum level: per cubic centimetre, **Gleason pattern 3 ≈ benign tissue**, whereas **Gleason pattern 4 contributes several-fold more serum PSA** than pattern 3, and more still than benign tissue.
- Without cancer, PSA varies with **age, race and prostate volume**.

Medications.

- Prostate-directed treatment (BPH or cancer) lowers PSA by **reducing epithelial volume and PSA production per cell**.
- **5α-reductase inhibitors (5ARI)** — finasteride, dutasteride — **lower PSA by about 50% after 12 months** → "doubling rule": multiply measured PSA by 2.
- **Doubling overestimates PSA in the first 6 months and underestimates it after several years.**
- **Prostate Cancer Prevention Trial (PCPT)** and other data: **5ARI therapy substantially improves the screening performance of PSA.**

PSA kinetics and density.

PSA VELOCITY VERSUS PSA DENSITY

	PSA velocity	PSA density
Definition	Rate of change of PSA	Total PSA (ng/mL) ÷ prostate volume (mL)
Reference values	No BPH or cancer: 0.04 ng/mL/year; BPH, age 60–85: 0.07–0.27 ng/mL/year	Proposed biopsy cut-offs 0.08–0.15
Early detection	Minimal, if any, value — even in high-risk men; never an isolated biopsy trigger	Substantially more predictive than PSA ; proposed after negative or equivocal magnetic resonance imaging (MRI), with variable success
Active surveillance	Conflicting data on predicting grade reclassification; utility unclear	≤0.15 an early Johns Hopkins AS inclusion criterion; key predictor of reclassification in the multicentre Prostate Cancer Active Surveillance Study

- **Trap:** the 0.15 density threshold is widely quoted after a negative MRI, but **data specifically supporting it in that setting are lacking.**

Concept — PSA rises when glandular architecture breaks down, not when cells make more of it: *this is why cancer, BPH, inflammation and instrumentation all raise it, why it is organ- rather than cancer-specific, and why volume-adjusted density outperforms the raw value.*

Free PSA

- **5–45% of serum PSA is free** and enzymatically inactive.
- **Cancer-derived PSA escapes proteolytic processing** → more complexed to ACT → **cancer is associated with a lower percentage free PSA (%fPSA).**
- **US Food and Drug Administration (FDA) approval: total PSA 4–10 ng/mL with a negative DRE.**
- Pre-biopsy thresholds: 18% improved discrimination over total PSA; thresholds of 14–28% avoid a substantial share of unnecessary biopsies while keeping sensitivity high. In a prospective multi-institutional study (men 50–75, PSA 4–10, benign DRE), **%fPSA 25% detected 95% of cancers** with a higher area under the receiver-operating-characteristic curve (AUC) than total PSA.
- **No single established threshold; proposed values 15–25%.**
- **National Comprehensive Cancer Network (NCCN):** lower %fPSA → greater risk of high-grade cancer; consider %fPSA to improve specificity **when evaluating for repeat biopsy.**
- **Risk calculators — PCPT and Prostate Biopsy Collaborative Group (PBCG)** — incorporate fPSA into calibrated estimates for shared decision-making.
- **Fig. 154.4:** web-based PCPT risk calculator (University of Texas Health Science Center at San Antonio); nomograms output risk of high-grade and of any cancer.
- **5ARI lowers fPSA in parallel with total PSA → %fPSA is not significantly altered.**
- **Commonest clinical use: elevated PSA after a previous negative biopsy.** Rebiopsy triggers of 21% (low underdiagnosis) and 28% (95% sensitivity) proposed; European repeat-biopsy data conflict — one cohort outperformed PSA and PCA3, another performed near chance → need for other markers.
- **Limitations:** well-studied inverse association with any cancer, but **not validated for GG ≥2** as contemporary markers are; reports give **any-grade risk, which grossly overestimates clinically significant risk** → some institutions remove any-grade estimates from reports.

Concept — **Free PSA exploits a processing defect:** *cancer PSA leaks out unprocessed and binds ACT, so the free fraction falls. Being a ratio, it survives 5ARI; being validated against any cancer, it underperforms for GG ≥2.*

Free PSA Isoforms

- Free PSA comprises **three isoforms in roughly equal concentrations** — proPSA, BPH-associated (benign) PSA (BPSA) and intact PSA (iPSA) — each a candidate marker.

FREE PSA ISOFORMS COMPARED

	proPSA ([−2]proPSA)	Benign PSA (BPSA)	Intact PSA (iPSA)
Nature	Incompletely cleaved precursor	Mature PSA with internal cleavages	Unnicked free PSA subset
Zone	Peripheral zone	Transition zone	—
Association	Cancer ; correlates with Gleason score	BPH and prostate volume ; no prognostic value	Cancer (intact:free ratio)
Clinical use	Prostate Health Index (PHI)	Possibly in combination only	4Kscore

ProPSA

- **[−7]proPSA** = 237 aa of mature PSA + 7-aa pro-leader; hK2 cleaves the leader to activate PSA. **Incomplete cleavage → truncated [−5], [−4] and [−2]proPSA**, primarily peripheral zone.
- With cellular disruption these inactive forms circulate as free PSA and **may form the majority of circulating free PSA in cancer**.
- Truncated forms markedly elevated in cancer tissue; **[−2]proPSA (p2PSA)**, cleaved between leucine 5 and serine 6 of the propeptide, is the most promising serum marker.
- Biology unclear: **reduced PSA processing in cancer** → relative excess of proPSA, especially p2PSA; alternatively, proPSA may be produced by **benign tissue adjacent to cancer**.
- **Extensively validated before initial and repeat biopsy**: %p2PSA an independent predictor of cancer and of GS ≥7 (PSA 2–10); in the US National Cancer Institute (NCI) Early Detection Research Network (EDRN) validation, it **outperformed PSA and %fPSA** and correlated with rising Gleason score. Combined with fPSA and total PSA (tPSA) as the **PHI**.
- **Fig. 154.5**: proPSA released with 7-aa leader (APLILSR) → hK2 cleavage → active PSA → proteolysis to iPSA, or internal nicking to BPSA; partial leader cleavage (ILSR, SR) → [−4] and [−2]proPSA.
- **Fig. 154.6**: normal versus cancer — in cancer, **loss of tissue architecture permits a relative increase in bound PSA and proPSA in serum**.

Benign PSA

- Identical to mature PSA except for **internal cleavages at Lys145 and Lys182**; largely confined to the **transition zone**, highly expressed in BPH.
- **Tracks prostate volume; no capacity to separate indolent from aggressive disease**; alone unlikely to distinguish BPH from cancer, possibly useful in combination.

Intact PSA

- Further fPSA subset; **intact-to-free ratio may improve detection**. Some assays also capture proPSA, confounding its independent value. **Used mainly as a component of the 4Kscore**.

Concept — *The free fraction is not one molecule: where PSA is processed determines what it signals — peripheral-zone clipped proPSA points to cancer, transition-zone BPSA points to BPH.*

IsoPSA

- **Structure-based rather than concentration-based assay**: partitions cancer-linked PSA isoforms in an aqueous two-phase reagent system.

- **Output K = ratio of complexed PSA in the bottom phase to the top phase;** used as binary, or converted by logistic regression to **KR, the individual probability (0–100%) of GG $\geq$ 2 cancer.**
- **Higher sensitivity and specificity than total PSA** for high-grade versus low-grade or benign disease, including with **equivocal MRI.**
- **Unaffected by 5ARI or α -blockers.**
- One of **six commercially available biomarkers in the 2023 NCCN Prostate Cancer Early Detection guideline** for risk stratification before biopsy.

4Kscore Test

- **Four kallikreins — tPSA, fPSA, iPSA and hK2 —** plus age, prior biopsy status and DRE findings $\rightarrow$ **probability of GG $\geq$ 2 cancer on biopsy (0–100%). hK2 is the unique component;** it activates proPSA by cleaving the pro-leader.

PSA VERSUS HK2

	PSA (hK3)	hK2
Shared	About 80% amino acid homology; highly prostate specific; androgen regulated	
Benign epithelium	Intense expression	Minimal immunoreactivity
Cancer	Lower per cell	More intense; associated with grade
Regulation	Expression independent of each other , in tissue and serum $\rightarrow$ hK2 complementary to PSA	

- Serum hK2 associated with cancer; hK2 with fPSA:tPSA improved detection in tPSA 2–10; hK2 also **prognostic** — adverse pathology and biochemical recurrence.
- **Development: Göteborg arm of ERSPC (PSA $\geq$ 3.0) —** high discrimination. **External validation: Rotterdam arm of ERSPC —** about half of biopsies avoidable at the cost of missing about one in eight high-grade cancers; at prostatectomy, the kallikreins independently predicted aggressive pathology.
- **Caveat: initial cohorts had lower GG $\geq$ 2 prevalence** than contemporary biomarker-tested populations; they nonetheless founded later validation.
- US biopsy-referral validation: **30–58% fewer biopsies across thresholds of 6–15%,** with few GG $\geq$ 2 cancers missed. Meta-analysis: **all four markers contribute;** adding the 4Kscore raises clinical-model AUC by about 0.11.
- **Reported cut-off 7.5%: sensitivity 94%, negative predictive value (NPV) 95%** across two large prospective validations.
- **Unique: low long-term risk of metastasis or prostate cancer death with a 4Kscore $\leq$ 7.5%** (two large observational studies).
- One of the **six NCCN 2023 biomarkers.**

Prostate Health Index

- **PHI = (p2PSA $\div$ fPSA) $\times$ $\sqrt$ tPSA;** single score ($\geq$ 0) associated with likelihood of cancer on biopsy.
- Multicentre study, PSA 2–10 with normal DRE: **PHI outperformed each of its components;** subsequent literature consistently better than PSA-based models.
- **GG $\geq$ 2 (since 2015):** better than PSA plus age and than PSA density in initial biopsy. Proposed thresholds: **24** (high sensitivity, about 30% of unnecessary biopsies avoided) and **28.6** (90% sensitivity, about 30% avoided).
- **Limitations: no single validated threshold;** reports give value ranges against **any-grade** cancer, not GG $\geq$ 2.

- One of the **six NCCN 2023 biomarkers**.

Concept — *Every blood test refines PSA by resolving its molecular forms — free fraction, clipped proPSA, intact PSA, hK2, isoform structure — because the forms, not the total, carry the cancer signal.*

URINE-BASED BIOMARKERS

PCA3

- Urine is easy to collect and consistently carries shed prostate cells. First urinary marker, identified 1999 by differential display on **chromosome 9q21–22**; named **prostate cancer antigen 3 (PCA3)** — a **long non-coding RNA (lncRNA)** expressed in cancer far above benign tissue.
- Assays: **reverse transcription polymerase chain reaction (RT-PCR) → transcription-mediated amplification (TMA)**, with better sensitivity and quantitation.
- **Requires urine collected after an attentive DRE; score = (urine PCA3 mRNA ÷ urine KLK3 mRNA) × 1000.**
- **Independent of prostate size** — the contrast with PSA.
- Thresholds most used: **10, 25, 35**; above 35, markedly higher biopsy detection than below. **The appropriate threshold depends on the population, especially previous biopsy status.**
- Nomograms: PCA3 improves a base clinical model (proposed cut-off 17); **updated PCPT calculator includes PCA3** as a continuous variable, outperforming the base PCPT model (see Fig. 154.4).
- **Agency for Healthcare Research and Quality** review: threshold 25 gives moderate sensitivity and specificity; PCA3 appears superior to total PSA, but **no evidence yet of better health outcomes.**
- **FDA approved (United States) and Conformité Européenne (CE) marked (Europe) for men with a previous negative biopsy.**
- **No longer available as an isolated test**; a key component of two NCCN urine assays (EPI and MPS).

SelectMDx

- **RT-PCR of post-DRE urinary mRNA of distal-less homeobox 1 (DLX1) and homeobox C6 (HOXC6)**, combined with age, DRE, PSA and **prostate volume** → probability of any cancer and of GG ≥2.
- Derived from three urinary markers (DLX1, HOXC6, Tudor domain-containing protein 1, TDRD1); the mRNA added modestly to a strong clinical model. **Over a quarter of the validation cohort had PSA ≥10** — unrepresentative of the usual biomarker population.
- Biopsy-naïve validation: **high sensitivity (about 93%) and NPV (about 95%) for GG ≥2**, maintained for PSA ≤10.
- **Limitation: needs prostate volume**, available only after transrectal ultrasound (TRUS) or MRI; without it, sensitivity, specificity and NPV all fall.
- One of the **six NCCN 2023 biomarkers**.

ExoDx Prostate IntelliScore

- ExoDx Prostate IntelliScore (EPI): **exosomal RNA of PCA3, ERG (ETS transcription factor) and SAM pointed domain-containing ETS transcription factor (SPDEF)** → score for GG ≥2.
- **Exosomes**: small vesicles shed into body fluids containing DNA, RNA and protein → urinary markers measurable **without a DRE**. Early work detected PCA3 and TMPRSS2:ERG in urinary exosomes.
- Predecessor **EXO106 (PCA3 + ERG)**: threshold 10, high sensitivity and NPV in PSA 2–10; adding clinical variables improved discrimination.

- EPI uses **no clinical factors**. **Initial biopsy, PSA 2–20: threshold 15.6** → about 92% sensitivity, NPV about 91%; about one-third of unnecessary biopsies avoided, about 8% of GG ≥ 2 missed; replicated in a second biopsy-naïve cohort.
- **Repeat biopsy**: 15.6 lost sensitivity; **threshold 20 kept sensitivity with better specificity and NPV** → thresholds are population specific.
- One of the **six NCCN 2023 biomarkers**.

MyProstateScore

- MyProstateScore (MPS; formerly Mi-Prostate Score, MiPS): **urinary PCA3 + TMPRSS2:ERG (T2:ERG) + serum PSA** → continuous score **0–100** for GG ≥ 2 .
- **TMPRSS2:ERG (landmark discovery, 2005)**: fusion of the 5′ untranslated region of the **androgen-regulated** transmembrane protease, serine 2 (TMPRSS2) gene with the E26 transformation-specific (ETS) transcription factors **ERG or ETS variant 1 (ETV1)**; **nearly 100% specific for prostate cancer**.
- Detectable in **post-DRE urine** (2008); TMA assay (2011) outperformed PSA; higher urinary score at prostatectomy → larger, higher-grade, non-organ-confined tumours.
- **Present in about half of tumours but about 75% of men with cancer** (multifocality) → **best combined with PCA3**; the combination roughly doubled specificity for GG ≥ 2 (17% to 33%) while holding sensitivity at 95%.
- Locked MPS model: **better discrimination for GG ≥ 2 than PSA or the PCPT calculator**; across thresholds missed $\leq 2.3\%$ of GG ≥ 2 cancers.
- **Initial biopsy: threshold 10** → 97% sensitivity, 98% NPV; about one-third of unnecessary biopsies avoided, 3% of GG ≥ 2 missed; confirmed in NCCN-eligible men, **Black men and men with a suspicious DRE**.
- **Repeat biopsy: threshold 15** (set for 90% target sensitivity) → 100% sensitivity and NPV on external validation.
- One of the **six NCCN 2023 biomarkers**.

miRNA

- Most genomic DNA does not encode protein. **MicroRNA (miRNA)**: small non-coding RNA (sncRNA), **usually 19–22 nucleotides**, regulating gene expression → potential diagnostic and prognostic markers across cancers.
- **miR-141**: associated with cancer and aggressive disease. **Urine miR-187**: independent predictor of cancer at biopsy.
- Multi-sncRNA panels are promising but have **no peer-reviewed clinical data yet**.

THE SIX NCCN 2023 PRE-BIOPSY BIOMARKERS

Test	Specimen	Components	Output	Key threshold(s)	Distinguishing feature
IsoPSA	Blood	PSA isoform structure (K)	KR: % risk GG ≥ 2	—	Unaffected by 5ARI or α -blockers
4Kscore	Blood	tPSA, fPSA, iPSA, hK2 + age, prior biopsy, DRE	% risk GG ≥ 2	7.5%	Low long-term metastasis and death risk below 7.5%
PHI	Blood	p2PSA, fPSA, tPSA	Score ≥ 0 (reported against any-grade cancer)	24 or 28.6 proposed; none validated	$(p2PSA \div fPSA) \times \sqrt{tPSA}$
SelectMDx	Post-DRE urine	DLX1, HOXC6 + age, DRE, PSA, volume	Risk of any cancer and GG ≥ 2	—	Needs prostate volume (TRUS or MRI)
EPI	Urine exosomes	PCA3, ERG, SPDEF	Score, GG ≥ 2	15.6 initial; 20 repeat	No DRE; no clinical variables
MPS	Urine + serum	PCA3, T2:ERG + PSA	0–100, GG ≥ 2	10 initial; 15 repeat	Validated in Black men and suspicious DRE

- **Pattern:** three blood, three urine; all target **GG ≥ 2** except PHI reporting, which remains any-grade.
- **Trap:** thresholds are **setting specific** — the repeat-biopsy threshold is higher (EPI 20, MPS 15) than the initial-biopsy one (EPI 15.6, MPS 10); %fPSA and PCA3 are likewise used mainly at repeat biopsy.

Concept — *The urinary markers are cancer-specific transcripts whose limitation is coverage, not specificity: a nearly 100%-specific fusion present in only half of tumours must be combined, and cells must be shed by DRE unless exosomes carry the RNA.*

TISSUE-BASED BIOMARKERS**Epigenetic Modifications**

- **Epigenetic change alters gene expression without altering DNA sequence** — DNA methylation and histone acetylation.
- **CpG islands:** GC-rich promoter segments; altered methylation changes expression and contributes to carcinogenesis. Lifelong exposures (diet, stress) may shift methylation and cancer risk.
- Hypermethylated genes implicated: **glutathione-S-transferase π (GSTP1)**, adenomatous polyposis coli (APC), retinoic acid receptor $\beta 2$ (RAR $\beta 2$), RAS association domain family protein isoform A (RASSF1A).
- **Field ("halo") effect:** methylation is detectable in **histologically normal tissue adjacent to tumour, not in distant normal tissue** → methylation in a **negative biopsy** suggests occult cancer.
- APC methylation in negative biopsies predicted cancer at repeat biopsy with high sensitivity and NPV.
- **ConfirmMDx (MDxHealth):** quantitative methylation of **GSTP1, APC and RASSF1**; independent predictor of cancer on repeat biopsy; **NPV about 88–90%**; in Black men, NPV about 94% for GS ≥ 7 .

Concept — *Methylation markers turn a negative biopsy into evidence: because the field effect extends into benign tissue beside the tumour, a core that missed the cancer can still carry its signature — hence their role before repeat biopsy.*

Genomic Expression Profiles

- Multigene panels outperform any single gene; run on archival **formalin-fixed paraffin-embedded (FFPE)** tissue; precedent in chemotherapy-response classifiers for breast and colorectal cancer. Available as **laboratory-developed tests (LDT)**.

TISSUE GENOMIC CLASSIFIERS COMPARED

	Prolaris	Oncotype Dx Prostate	Decipher
Genes	31 cell cycle progression (CCP) + 15 housekeeping	12 cancer genes in 4 pathways + 5 housekeeping	22-marker classifier
Predicts	Progression and prostate cancer-specific death (conservatively managed); biochemical recurrence and metastasis after radical prostatectomy; outcome including cancer-specific mortality after external beam radiotherapy	Adverse pathology at prostatectomy = primary Gleason pattern 4 or pT3	Metastasis and cancer-specific mortality in high-risk disease and biochemical recurrence
Principal setting	Newly diagnosed: AS versus treatment	Newly diagnosed: AS versus treatment	Developed post-prostatectomy; now also newly diagnosed (biopsy)
Notes	Adds to Cancer of the Prostate Risk Assessment (CAPRA) score for recurrence after prostatectomy	Extensively validated for clinical outcome	Low, intermediate, high risk → rising 10-year post-prostatectomy metastasis; score ≥ 0.4 may benefit from postoperative radiotherapy (retrospective)

- **Post-Operative Radiation Therapy Outcomes Score (PORTOS):** 24-gene signature derived from the Decipher database to predict **response to adjuvant radiotherapy**.
- **Limitations — multifocality and heterogeneity:** distinct foci in one patient show discordant expression profiles; **scores from a low-grade focus do not predict an unsampled high-grade focus**; almost all evidence is **retrospective**.
- **Genomics in Michigan to Adjust Outcomes in Prostate Cancer (G-MAJOR):** ongoing prospective randomised trial of Decipher in decision-making for newly diagnosed favourable-risk cancer (NCT04396808).

Concept — *Tissue classifiers answer prognosis, not detection, and they can only describe the focus sampled; multifocal heterogeneity is their ceiling.*

Germline Genetics and Prostate Cancer

- **Up to 40–50% of prostate cancer risk may be familial or hereditary.** Familial cancer (15–20% of cases) includes strong family history without a detectable mutation.
- **An affected relative raises risk two- to fourfold.**
- **Single nucleotide polymorphisms (SNPs):** common variants, not necessarily functional; **more than four risk SNPs → 4.5-fold risk, tenfold with a family history.**
- **STHLM3:** the four 4Kscore kallikreins + **232 risk SNPs** + clinical variables → composite early-detection test for GS ≥ 7 (AUC 0.74).
- **Most heritable genes are DNA damage repair genes.**

GERMLINE PROSTATE CANCER RISK GENES

Gene	Function	Prostate cancer risk	Phenotype
BRCA2	Homologous recombination	4.5-fold under 65; about 8.6-fold across all ages	Younger diagnosis, higher grade, shorter survival
BRCA1	Homologous recombination	1.8-fold under 65	—
Lynch syndrome (MLH1, MSH2 , MSH6, PMS2)	Mismatch repair	MSH2: about tenfold relative risk	Also colorectal, liver, skin, upper tract cancers
CHEK2	Double-strand break repair	Odds ratio about 2 (unselected), about 3.4 (familial)	—
ATM	Double-strand break repair	Increased	Aggressive disease (with BRCA1/2)
HOXB13 G84E	Homeobox protein	4.5-fold (familial linkage)	No evidence of more aggressive cancer

- **Trap:** HOXB13 raises risk but not aggressiveness; **BRCA2 raises both.**
- **Next-generation sequencing (NGS):** single-gene (BRCA1/2) or, without a known familial mutation, **multigene panels** (BRCA1, BRCA2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM, others) — some genes of **unknown significance**. **NCCN focuses on BRCA testing;** carriers need tailored early detection and management.

Concept — *Hereditary prostate cancer is predominantly a failure of DNA damage repair, and the genes that matter most clinically — BRCA2 above all — are those that raise aggressiveness, not merely incidence.*

SUMMARY

- **Early detection while cancer is organ confined improves cure rates** and lowers morbidity and mortality.
- PSA revolutionised detection, but **stage migration and a changed natural history have outrun it; PSA derivatives improve sensitivity at the risk of specificity.**
- Blood, urine and tissue markers now improve detection and risk stratification; **panels will probably overcome multifocality and heterogeneity.**
- Translation requires proof of **clinical validity and clinical utility** — real impact on care.

KEY POINTS

- **PSA = hK3**, human kallikrein family; **PSA and hK2** both used in detection.
- **Organ specific, not cancer specific;** produced by prostatic epithelium; **half-life 2–3 days.**
- **Gram for gram, cancer produces less PSA than benign tissue;** serum rise reflects disrupted architecture; also BPH, prostatitis, recent manipulation.
- **About 70% of serum PSA bound** (ACT, A2M, API); **cancer → higher bound fraction, lower free PSA.**
- Pre-pro-PSA → [–7]proPSA (237 + 7 aa) → **hK2 cleavage → active PSA;** incomplete cleavage → [–2], [–4], [–5]proPSA, which augment detection.
- **5ARI reduce serum PSA by 50%.**
- **Circulating tumour cells (CTCs) and circulating tumour DNA** are promising marker sources, particularly in advanced disease.
- **PCA3:** urinary non-coding RNA; combined with the highly specific **TMPRSS2:ERG** fusion for greater accuracy.

- **Tissue gene-expression signatures** run on small FFPE samples add prognostic information beyond Gleason score.
- Susceptibility genes on several chromosomes; **moderate-penetrance genes for aggressive cancer: BRCA1, BRCA2, ATM** — all DNA damage repair.